The Landscape of Supernovae

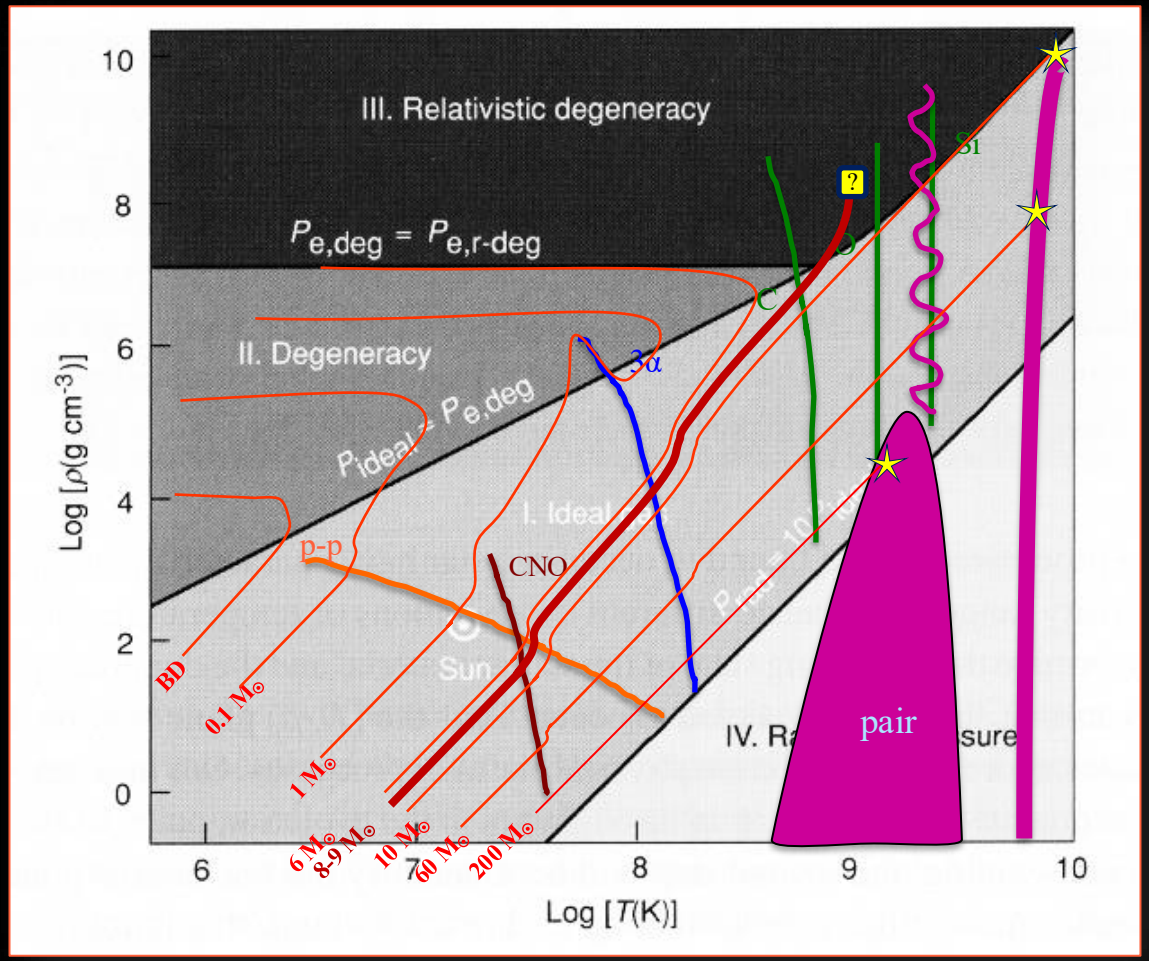
ASTR 400A – Theoretical Astrophysics
Fall 2026

Prof. Carl Fields
TA & GRA Brain Hsu (Guest Lecture)

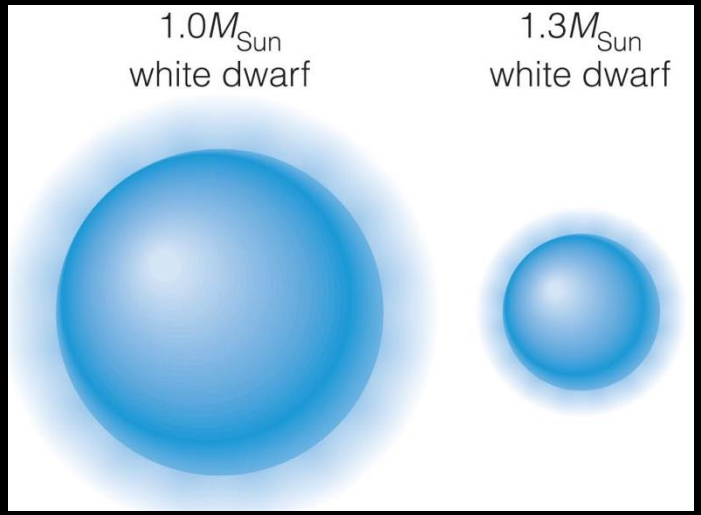
10/27/2026

Background: SN 1987A – NASA/ESA/CSA JWST/NIRCam

Core Conditions Dictate Final Fate



White Dwarfs

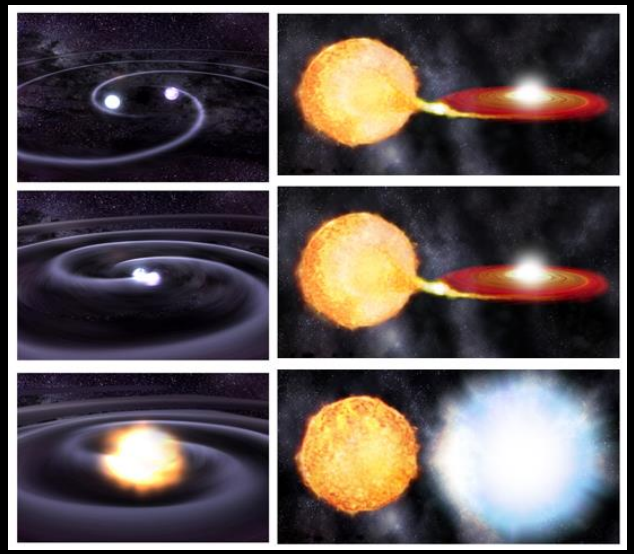




- Accreting WD continues to gain mass (and shrink in size), but not forever!
- Binary white dwarfs can also get close enough to tidally disrupt (making an accretion disk onto more massive WD) or merge into one.
- When it reaches a mass of 1.4 $M_{\odot}$, the entire white dwarf (mostly carbon) burns in a violent thermonuclear explosion.

White Dwarf Supernova

- WD Supernova completely burns and explodes the WD
- Nothing left behind (no neutron star or black hole)
- They are bright! About 10^{43} erg/s at peak
- Because all are explosions of WD with same mass (1.4 $M_{\odot}$), all WD supernovae have *roughly* the **same luminosity**
- Two scenarios:
  - Single degenerate with a MS/RG/He star
  - Double degenerate with another WD

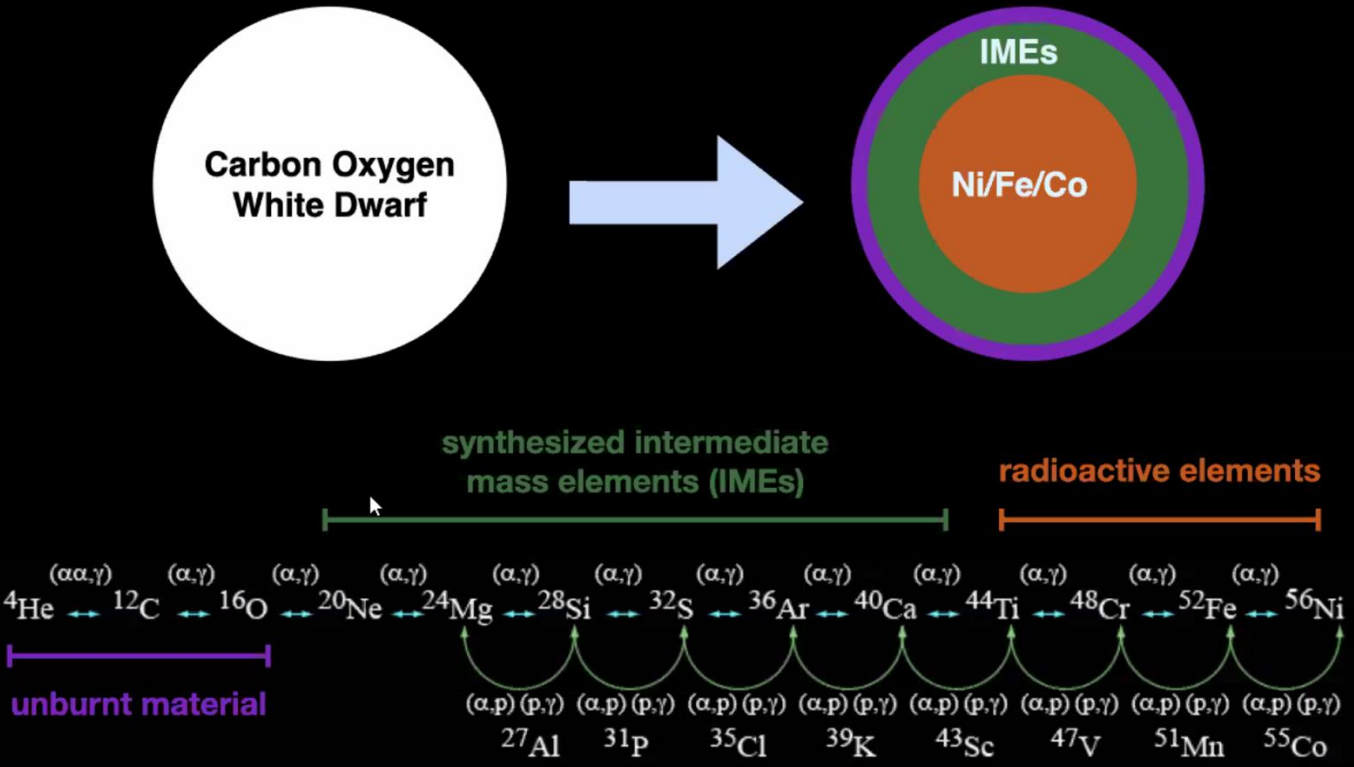


Thermonuclear Supernova

or

Type Ia Supernova

C/O WD and Thermonuclear Reactions



Type Ia SNe lose all shock-deposited heat due to adiabatic expansion (PdV work). They would be invisible if it were not for reheating of the ejecta by radioactive decay.

Radioactive decay:

$^{56}Ni \rightarrow {}^{56}Co \rightarrow {}^{56}Fe$

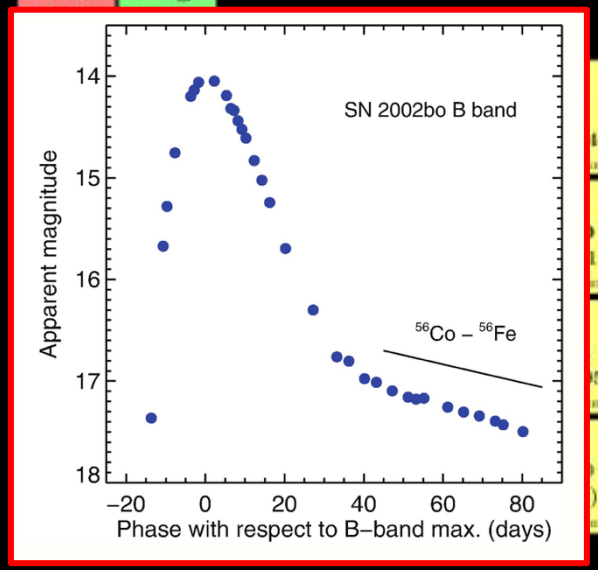


1 H 1.01 Hydrogen																
3 Li 6.94 Lithium	4 Be 9.01 Beryllium											5 B 10.81 Boron	6 C 12.01 Carbon	7 N 14.01 Nitrogen	8	
11 Na 22.99 Sodium	12 Mg 24.31 Magnesium											13 Al 26.98 Aluminum	14 Si 28.09 Silicon	15 P 30.97 Phosphorus	16	
19 K 39.10 Potassium	20 Ca 40.08 Calcium	21 Sc 44.96 Scandium	22 Ti 47.87 Titanium	23 V 50.94 Vanadium	24 Cr 52.00 Chromium	25 Mn 54.94 Manganese	26 Fe 55.85 Iron	27 Co 58.93 Cobalt	28 Ni 58.69 Nickel	29 Cu 63.55 Copper	30 Zn 65.39 Zinc	31 Ga 69.72 Gallium	32 Ge 72.61 Germanium	33 As 74.92 Arsenic	34	
37 Rb 85.47 Rubidium	38 Sr 87.62 Strontium	39 Y 88.91 Yttrium	40 Zr 91.22 Zirconium	41 Nb 92.91 Niobium	42 Mo 95.94 Molybdenum	43 Tc (98) Technetium	44 Ru 101.07 Ruthenium	45 Rh 102.91 Rhodium	46 Pd 106.42 Palladium	47 Ag 107.87 Silver	48 Cd 112.41 Cadmium	49 In 114.82 Indium	50 Sn 118.71 Tin	51 Sb 121.76 Antimony	52	
55 Cs 132.91 Caesium	56 Ba 137.33 Barium	57-70 *	71 Lu 174.97 Lutetium	72 Hf 178.49 Hafnium	73 Ta 180.95 Tantalum	74 W 183.84 Tungsten	75 Re 186.21 Rhenium	76 Os 190.23 Osmium	77 Ir 192.22 Iridium	78 Pt 195.08 Platinum	79 Au 196.97 Gold	80 Hg 200.59 Mercury	81 Tl 204.38 Thallium	82 Pb 207.2 Lead	83 Bi 208.98 Bismuth	84
87 Fr (223) Francium	88 Ra (226) Radium	89-102 **	103 Lr (262) Lawrencium	104 Rf (261) Rutherfordium	105 Db (262) Dubnium	106 Sg (266) Seaborgium	107 Bh (264) Bohrium	108 Hs (269) Hassium	109 Mt (268) Meitnerium	110 Ds (271) Darmstadtium	111 Rg (272) Roentgenium	112 Cn (283) Copernicium	113 Uut [286] Ununtrium	114 Fl (287) Flerovium	115 Uup [288] Ununpentium	

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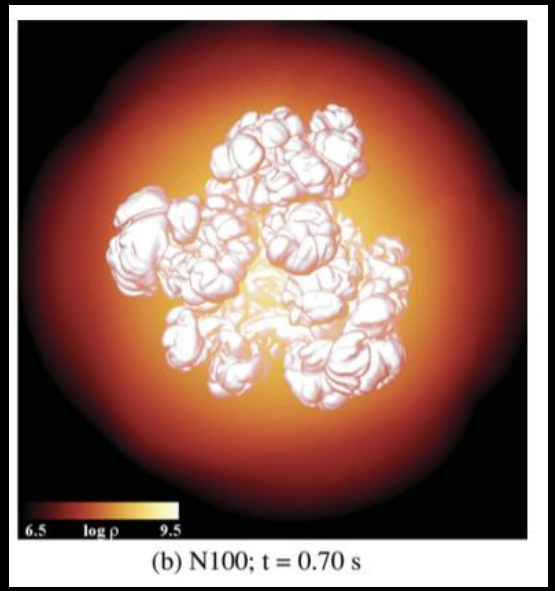


^{56}Ni half-life = 6.1 d
decay time = $t_{1/2}/\ln(2)$ = 8.8 days
so $L = L_0\, e^{-8.8}$ --- declines at 1/8.8 mag/day

^{56}Co half-life = 77.1
decay time = $t_{1/2}/\ln(2)$ = 111 days
so $L = L_0\, e^{-111}$ --- declines at 1/111 mag/day

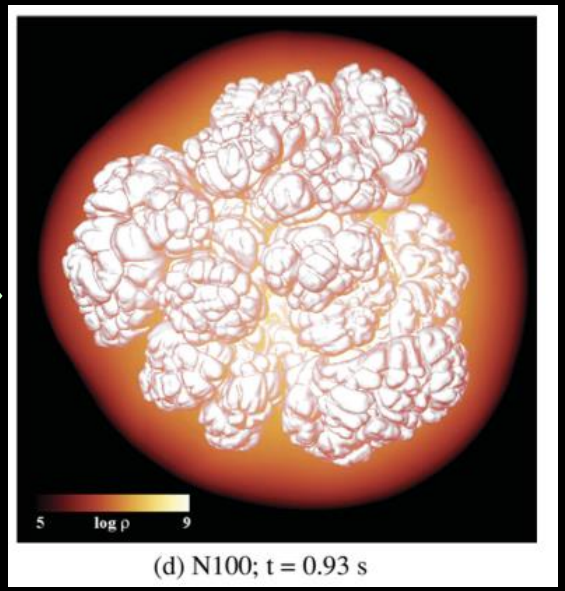
Mechanism 1: Deflagration-to-Detonation Transition

Seitenzahl et al. (2013)



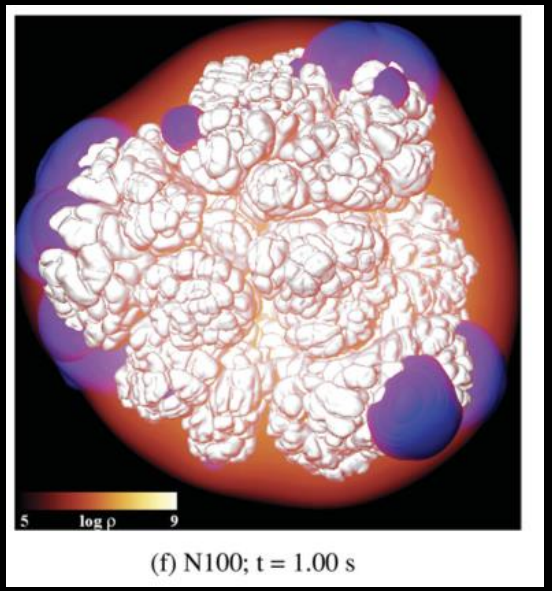
(b) N100; t = 0.70 s

Central deflagration



(d) N100; t = 0.93 s

Transition to detonation



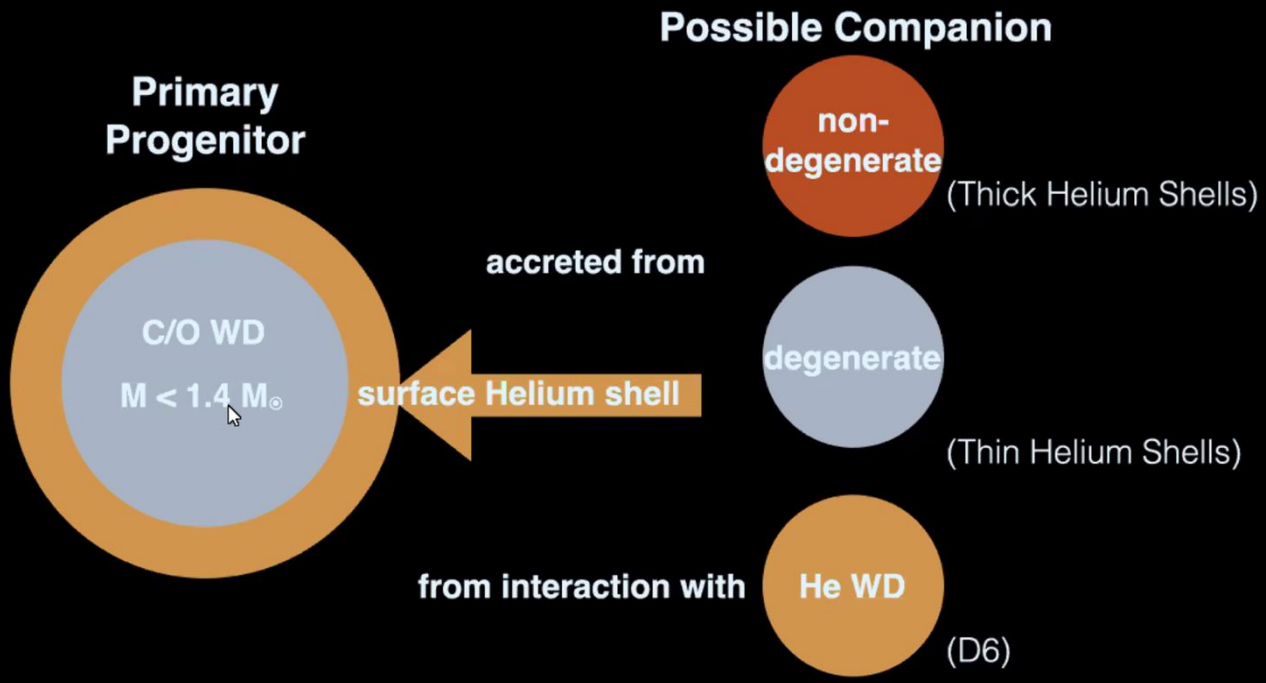
(f) N100; t = 1.00 s

Full detonation

Deflagration: subsonic burning; Detonation: supersonic burning

This mechanism is also often referred to as the Delayed-Detonation Model

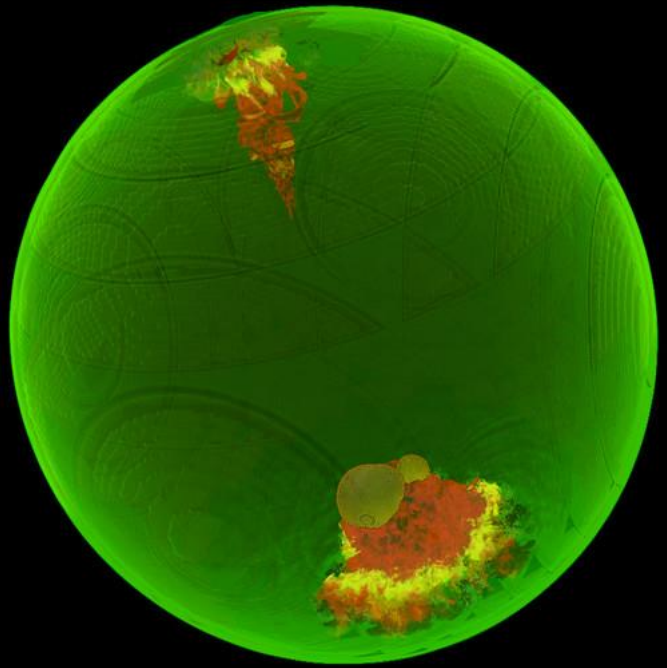
Mechanism 2: Double Detonation



Double Detonation Scenario: detonation of the surface helium leads to a detonation of the carbon-oxygen core

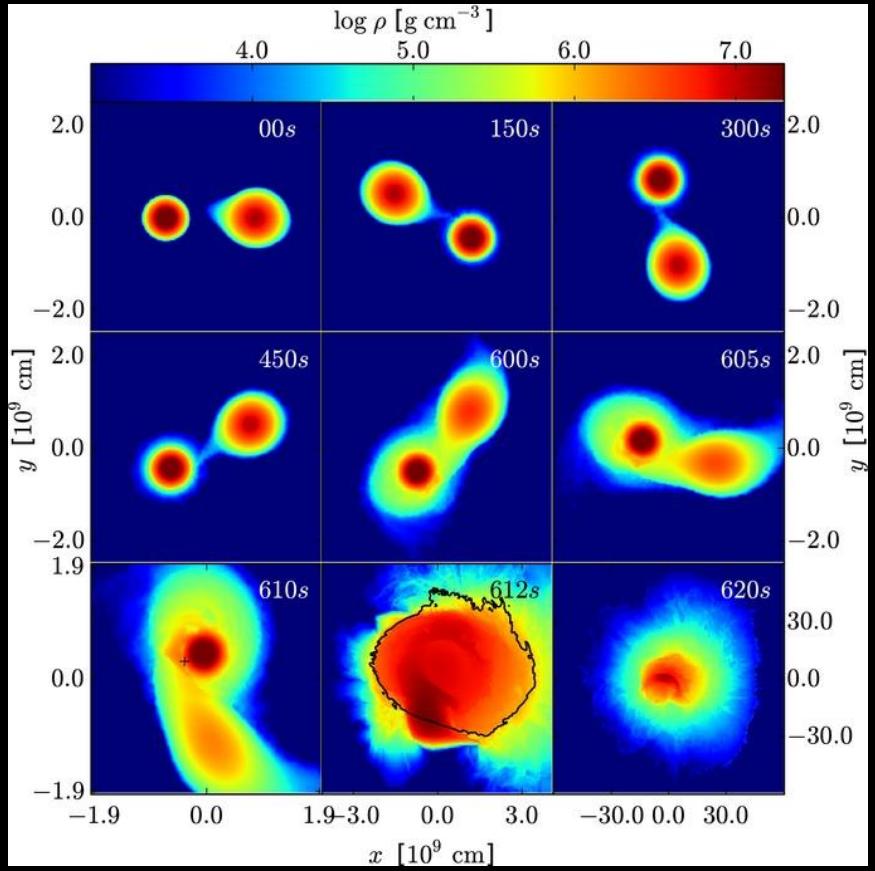
Mechanism 3: Gravitationally Confined Detonation

- Deflagration occurs off-center
- A hot bubble consists of burning ash expands and rises toward the surface
- Burning front expands laterally around the surface
- The fronts collide at the other side of the surface and compress unburned material
- This detonates the WD



Scenario 4: Violent Mergers

- Binary WDs in-spiral and merge
- Secondary WD is disrupted and accreted onto the primary
- Detonation occurs upon impact of the disrupted secondary

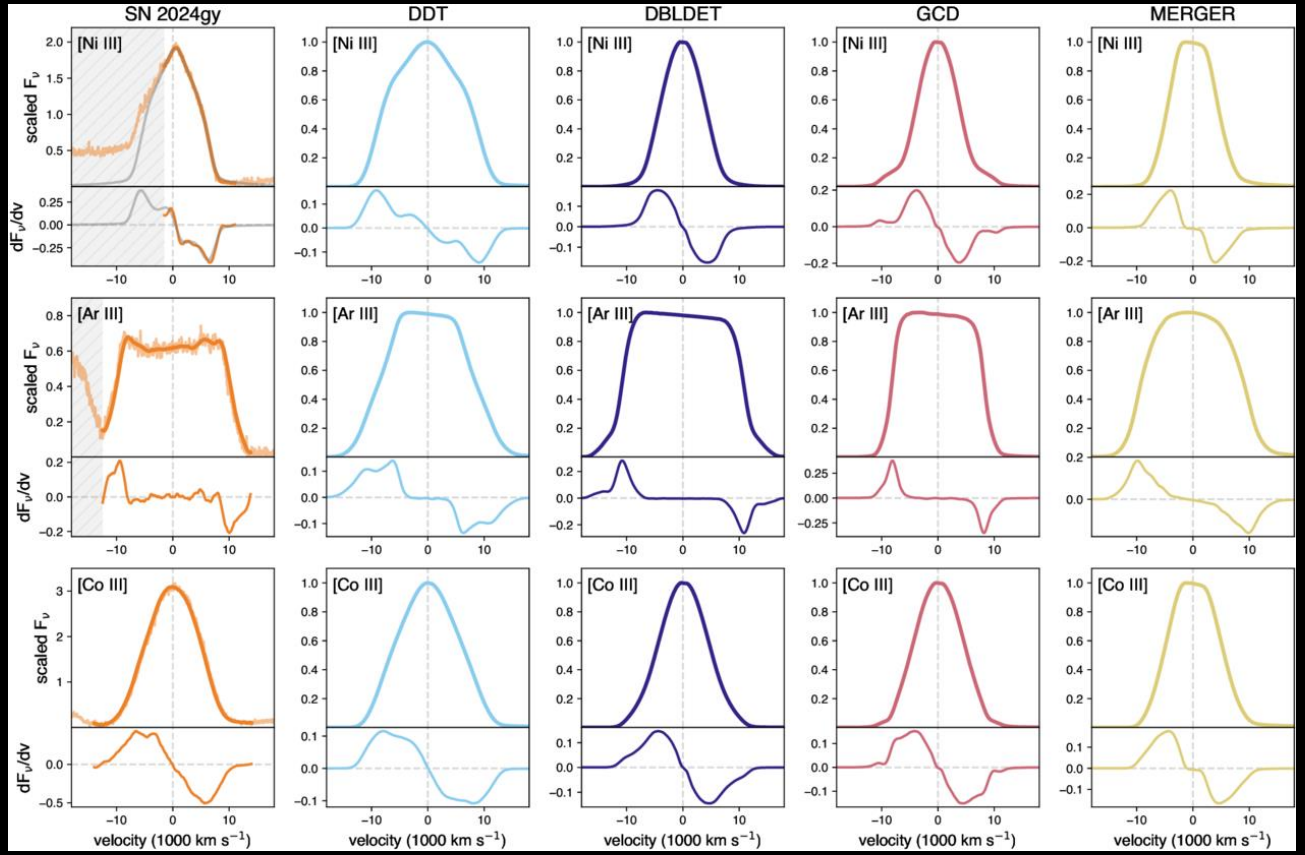


Pakmor et al. (2012)

How do we distinguish between mechanisms?

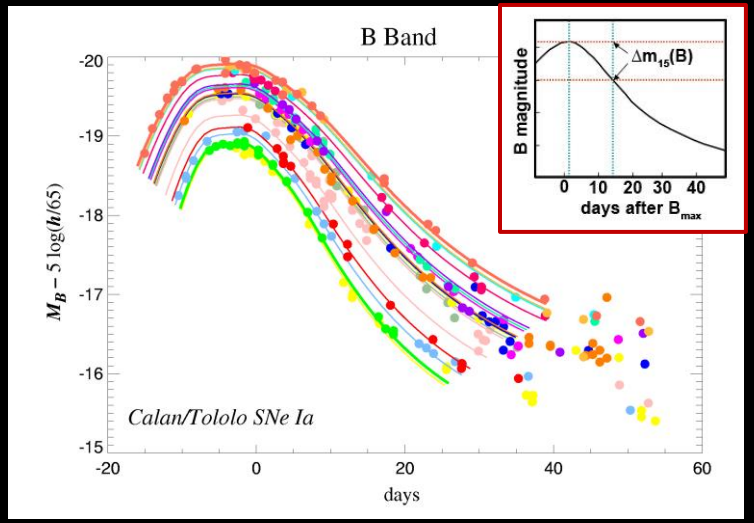
Nebular spectra with JWST!

Different atomic/ionic lines show different profile shapes depending on the explosion mechanism.



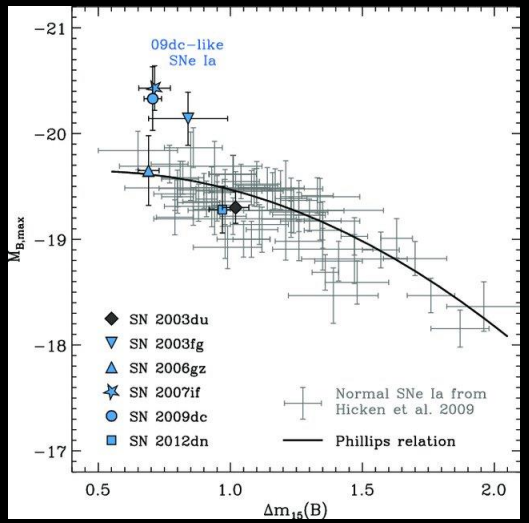
Kwok et al. (2026)

Type Ia SN Cosmology



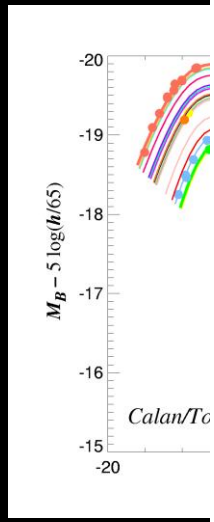
Type Ia SNe are not standard candles, but they are "standardizable" candles.

More luminous = slower decay
Less luminous = faster decay



Luminosity–decline rate relation, a.k.a. the "Phillips relation"

Nobel Prize in Physics 2011





© The Nobel Foundation. Photo: U. Montan

Saul Perlmutter

Prize share: 1/2



© The Nobel Foundation. Photo: U. Montan

Brian P. Schmidt

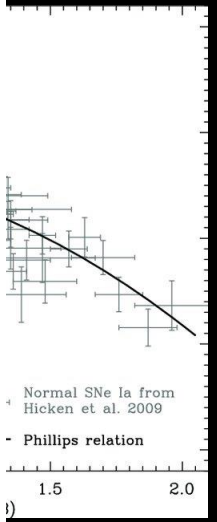
Prize share: 1/4



© The Nobel Foundation. Photo: U. Montan

Adam G. Riess

Prize share: 1/4



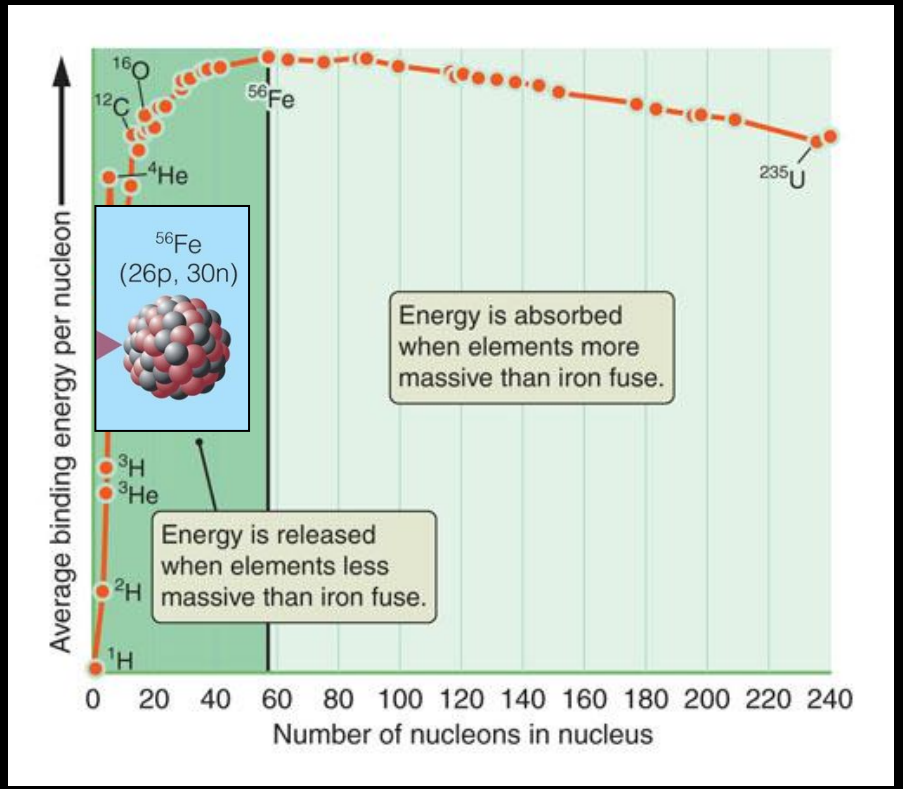
The Nobel Prize in Physics 2011 was divided, one half awarded to Saul Perlmutter, the other half jointly to Brian P. Schmidt and Adam G. Riess "for the discovery of the accelerating expansion of the Universe through observations of distant supernovae"

Type Ia SN
candles, b
"standard
More lum
Less lumi

cline rate
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Nuclear Burning in Massive Stars

- Cores of high-mass stars are hot enough to have fusion reactions beyond carbon.
- Can produce all elements up to iron (Fe)
- Even a high-mass star can't burn iron, because it doesn't produce energy.



Fe core builds up in mass, as gravity gets stronger and stronger…

Advance Burning Stages in Massive Stars

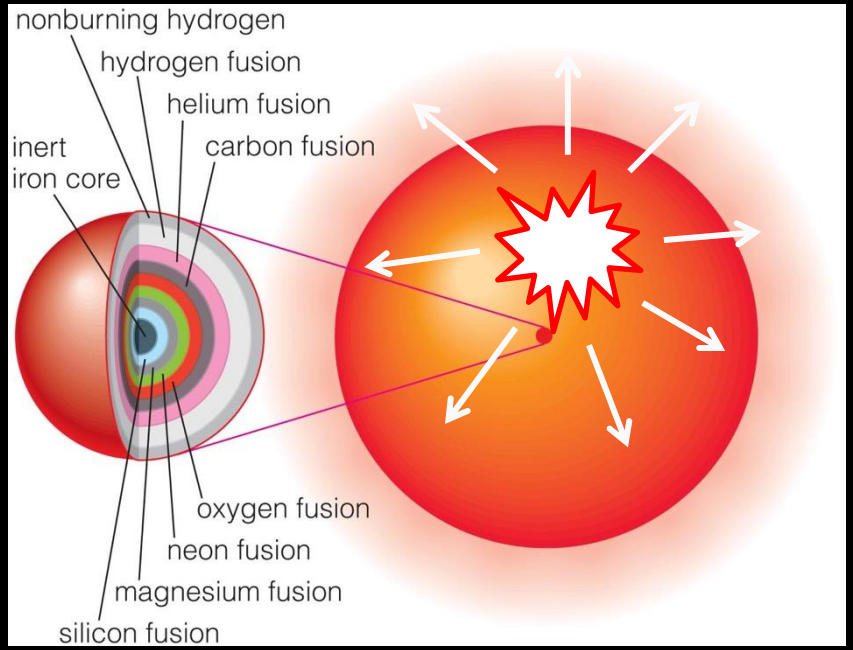
Advanced core burning stages are fast – envelopes can't adjust (t_{diff}).

Burning times and T (20 $M_{\odot}$):

H ~ 1e7 yr; 1e7 K
He ~ 0.5-1e6 yr; 1e8 K
C ~ 1000 yr; 6e8 K
Ne ~ 1 yr; 1.5e9 K
O ~ ≤1 yr; 2e9 K
Si ~ 1 day; 3.5e9 K

} Heavy neutron excess, neutrino cooling

Core convergence… next burning sequences starts when shell has added enough fuel (roughly at M_{Ch}).

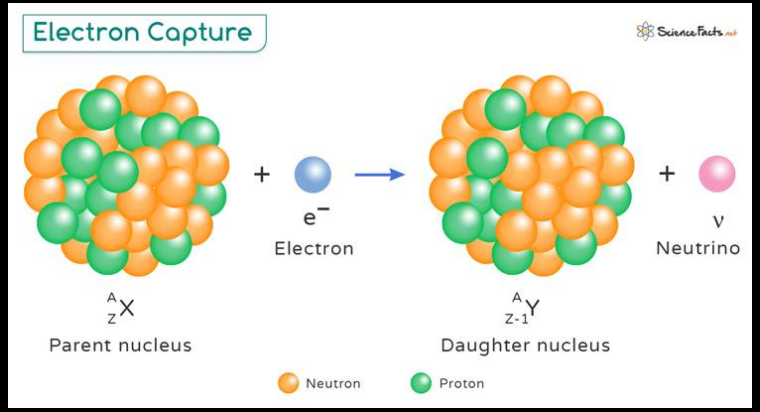


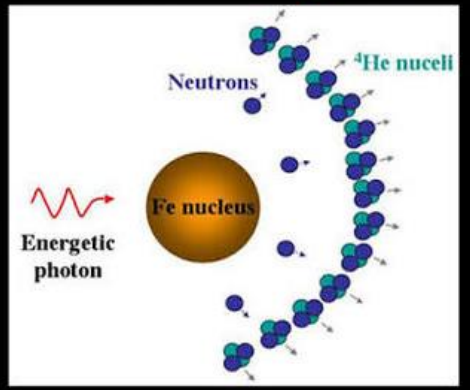
Iron Core Collapse

As with C/O WDs, inert Fe cores also have a corresponding $M_{Ch} \approx 1.26\ M_{\odot}$.

Two processes accelerate the core collapse:

1. **Electron captures:**
   - At high density, free electrons can be captured via inverse beta decay - core becomes more neutron rich - *neutronization*.
   - This reduces the free electron number and therefore M_{Ch}.
2. **Photo-disintegration:**
   - At ~10^{10} K, gamma rays can break up heavy nuclei into lighter ones
     $^{56}Fe + \gamma \Leftrightarrow 13(^{4}He) + 4n$ and $^{4}He + \gamma \Leftrightarrow 2p + 2n$
   - This removes pressure support from the Fe core, potentially triggering the Fe core collapse.



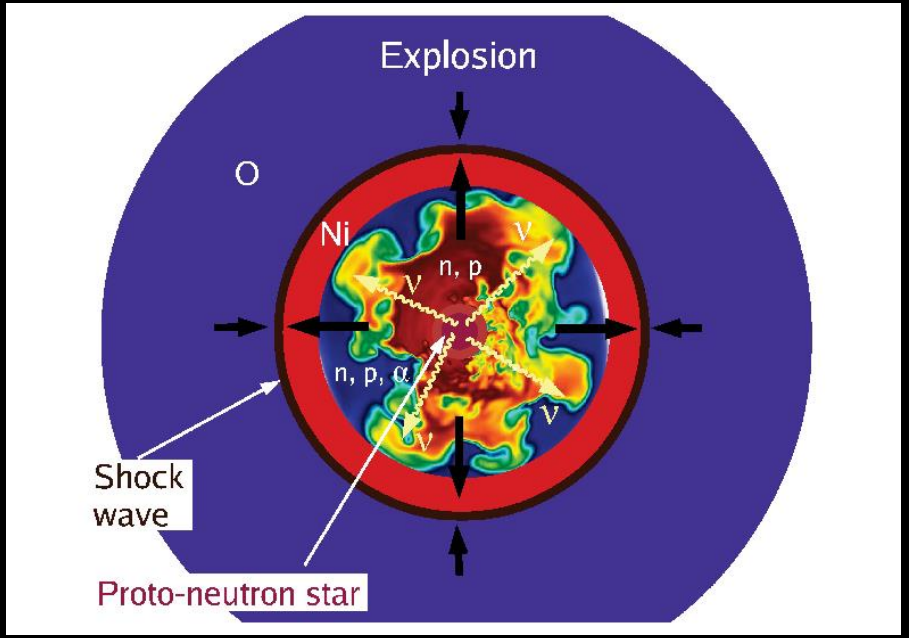


Iron Core Collapse

- The collapse is extremely rapid, ~10 ms (recall our dynamical timescale and typical core density $\rho \sim 10^{10}$ g cm^{-3}).
- Photo-disintegration and electron captures continue due to increases in photon energies/density and lead to further neutronization.
- The temperature and pressures rise, but not enough to halt collapse until nuclear densities are reached $\rho \sim 10^{14}$ g cm^{-3}).
- The composition becomes predominantly neutrons, modifying the equation of state, the gas becomes nearly *incompressible* and the proto-neutron star halts collapse at a radius of about 20 km.

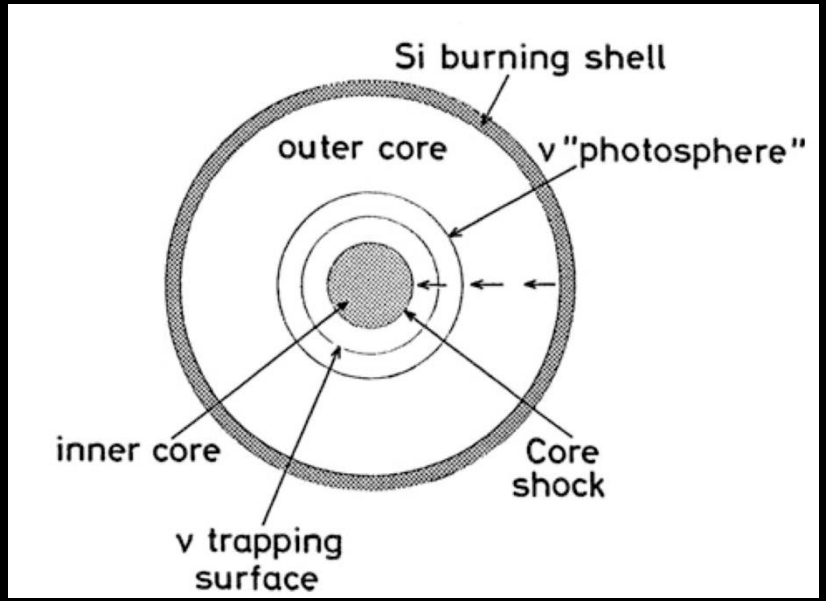
The Delayed Neutrino-heating Mechanism

- At a few ~3×10^{14} g cm^{-3}, neutrons become degenerate and the inner core like a spring causing the **core bounce**.
- The velocity of the inner core is reversed due to the bounce and travels towards still infalling material of the outer core.
- These material impact each other, the supersonic material creates a shock wave that steepens as it moves outward into lower density.
- Photo-disintegration and electron captures take away most of the energy from the shock, and the shock front stalls.
- If the shock is not revived, the star doesn't explode and becomes a **failed supernova**.



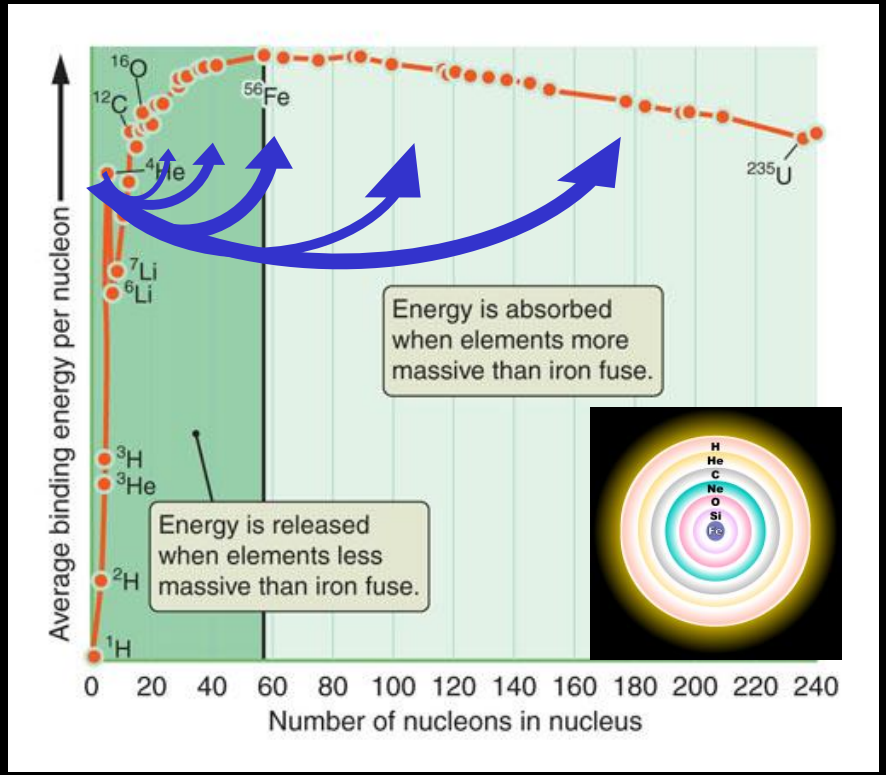
The Explosion Mechanism

- The core becomes opaque to neutrinos and forms a ***neutrinosphere*** - outer layers of the core where the density is low enough for neutrinos to escape. Within this radius neutrinos have to scatter many times before escaping.
- **Neutrino heating** - neutrinos able to diffuse out of the core heat the region where the shock has already passed (the post shock region) - from about 30 km to 100-300 km.
- This neutrino-heated region becomes convective and provides a way to convert thermal energy deposited by neutrinos into kinetic energy potentially leading to revival of stalled shock overcoming ram pressure to lead to ***explosion***.



Explosive Nucleosynthesis

- Most of the Fe goes into the proto-NS.
- Infalling Si, Mg, Ne, O may also disintegrate into He nuclei.
- NSE fuses these He nuclei to heavier elements.
- Expansion cools the ejecta, and heavier elements “freeze out” before they are destroyed.

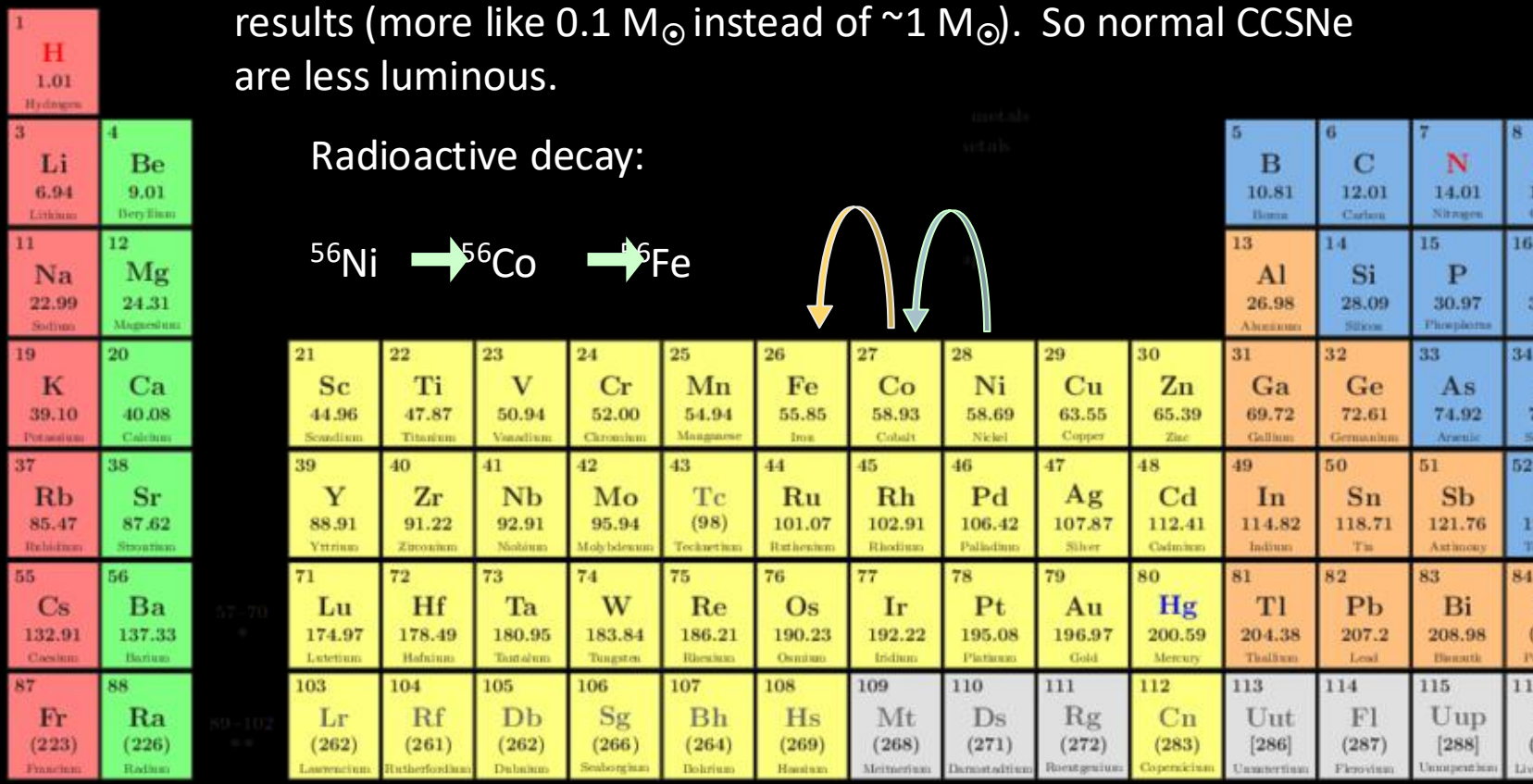


Note that the Fe produced in the star’s core will never escape (goes into the NS or BH). Massive stars are not primary source of Fe.

As in Type Ia SNe, NSE burning in CCSNe makes a lot of Fe-peak elements, especially ^{56}Ni. However, much smaller 56Ni mass results (more like 0.1 $M_\odot$ instead of ~1 $M_\odot$). So normal CCSNe are less luminous.

Radioactive decay:

^{56}Ni → ^{56}Co → ^{56}Fe



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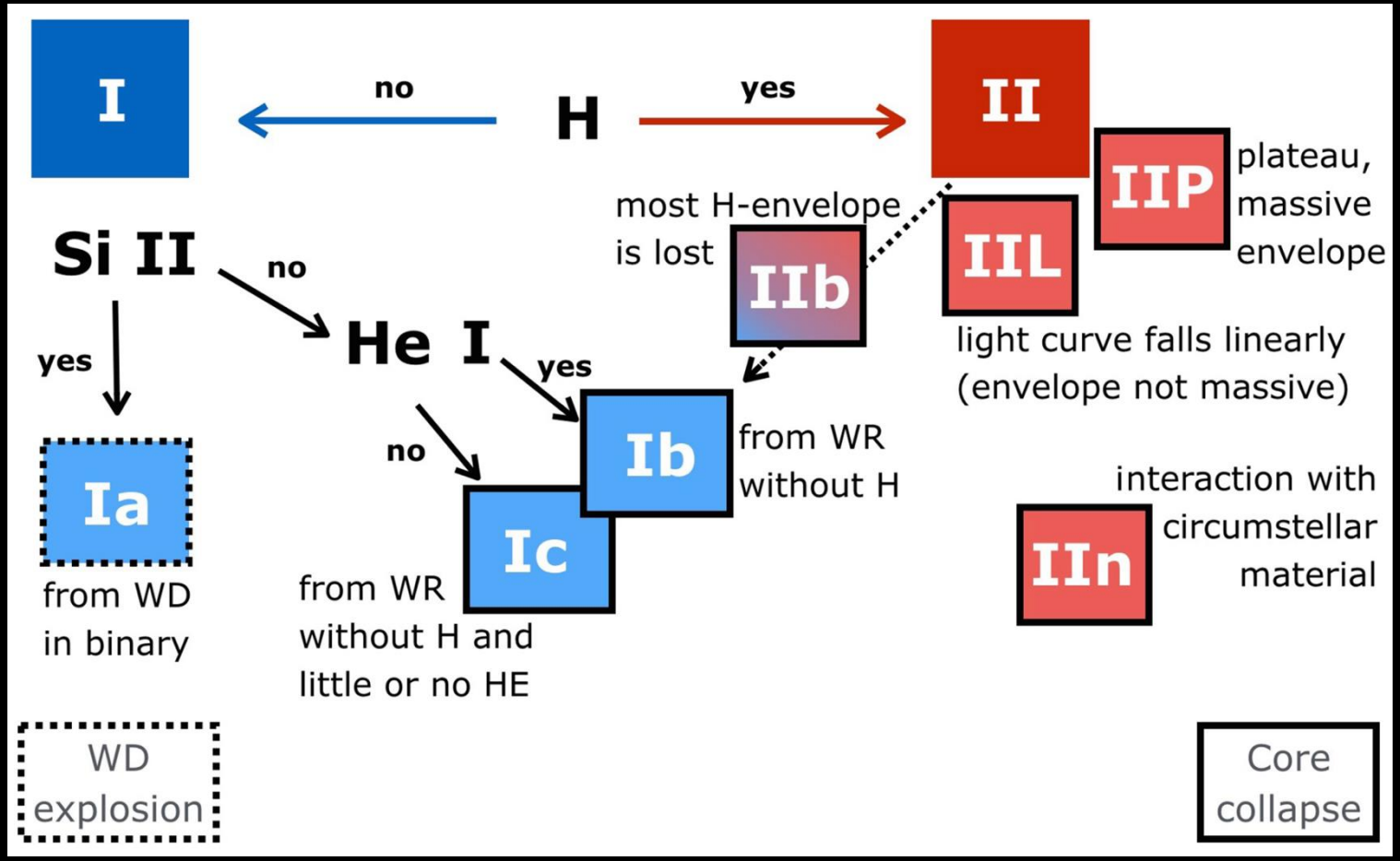
Radioactive decay:

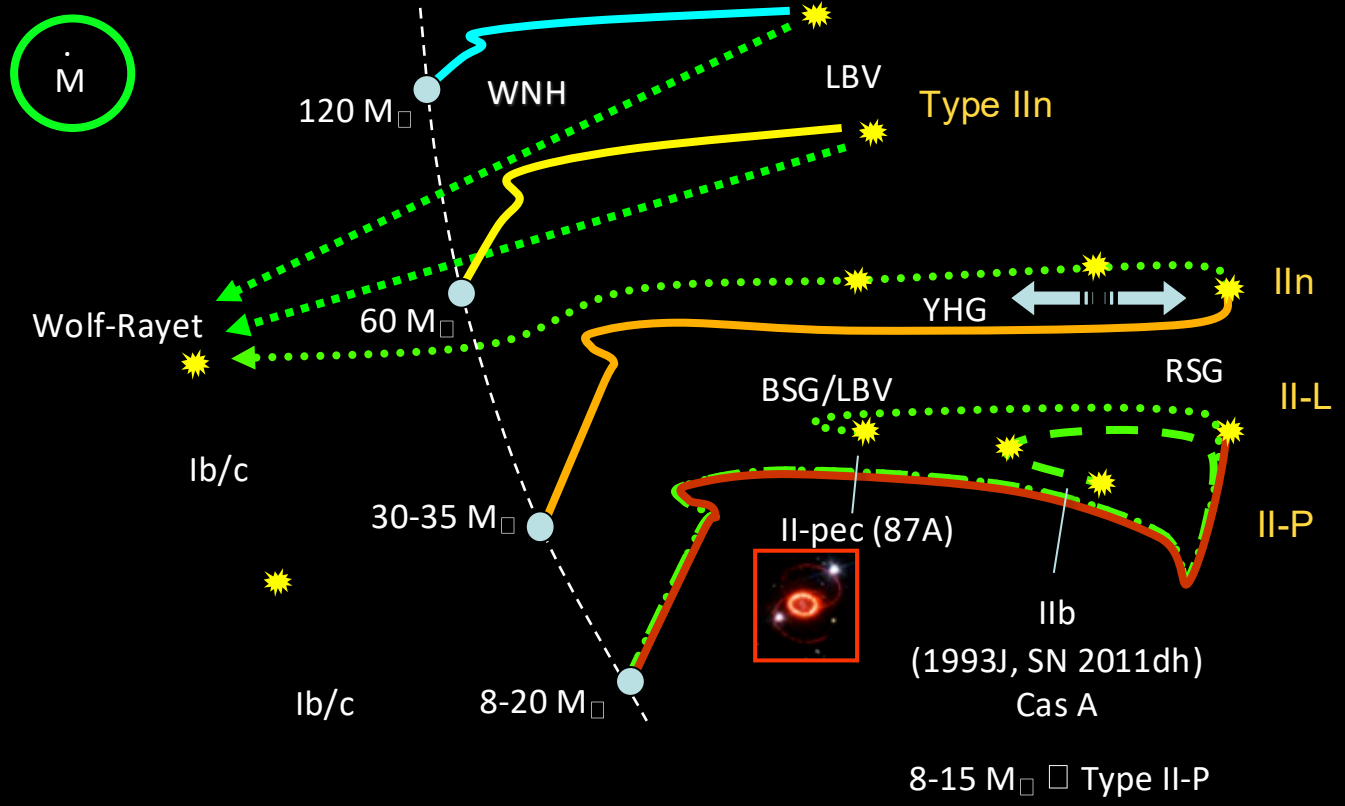
$^{56}Ni \rightarrow {}^{56}Co \rightarrow {}^{56}Fe$



NSE burning only lasts a short time as the shock passes through layers outside Fe core. Only a mall fraction of star is burned, so primary nucleosynthesis yield is intermediate-mass elements like O, Mg, S, etc. plus small mass of r-process elements.

Supernova classification is purely observational





But something like 50% or more of Massive stars are in close binaries…

END FATES of MASSIVE STARS:
What type of supernova from which type of star?

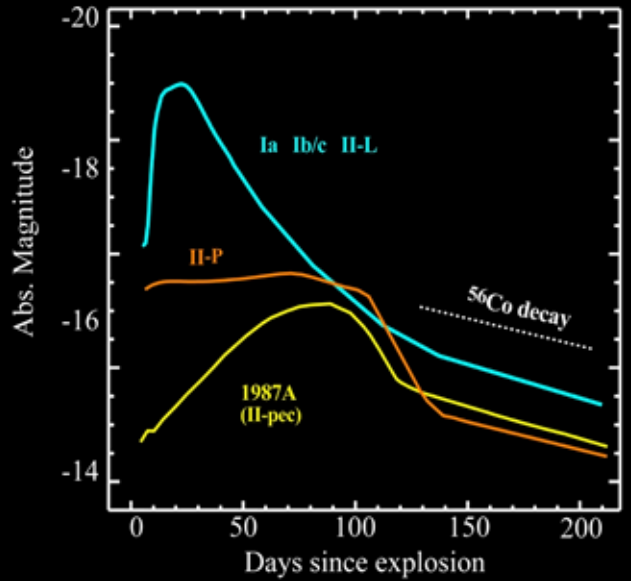
Light Curves

Type Ib/Ic: core-collapse SNe with no H. a.k.a.: "stripped-envelope SNe"

- WR star progenitors (small R).
- Light curves look like SNe Ia (but fainter)
- Powered entirely by radioactive decay.

Type II-P: progenitors with H envelopes. RSGs produce light curve with a "plateau". Plateau is powered by shock heated envelope at first, then radioactive decay toward end. Plateau ends when whole H envelope recombines and becomes optically thin. Plateau time is approximately the diffusion time.

Type II SNe with more compact BSG progenitors (like 1987A) are fainter at first.



$$t_{\rm diff} \approx \frac{\tau R}{c} = \frac{R^2}{Lc} = \frac{\kappa \rho R^2}{c}$$

Light Curves

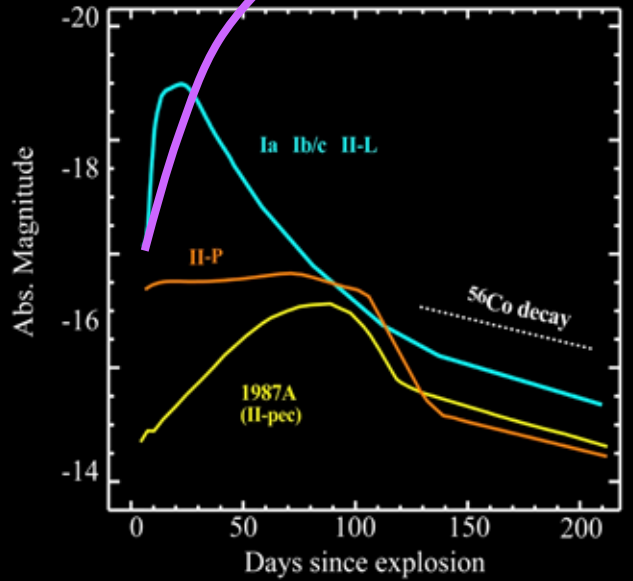
Pair Instability SN

makes ~10 $M_\odot$ of ^{56}Ni

This makes it a “Super-Luminous SN”

(about 10x brighter than brightest Type Ia SNe, which are brighter than core collapse SNe)

Since star that exploded is very massive (>100 $M_\odot$), diffusion time is longer and PISNe take a longer time to brighten and fade.



$$t_{\rm diff} \approx \frac{\tau R}{c} = \frac{R^2}{Lc} = \frac{\kappa \rho R^2}{c}$$

In-class Assignment 16

Work on ICA 16 here with groups per usual.

Discuss conceptual questions together and prepare answers to share at the end of class.

Choose someone that will report out the group's responses ahead of time!

After Class: Due: Submitted to D2L by EoD

Note: The goal of ICAs are to use plots produced in the notebook for discussion and interpretation of results presented in lecture in groups *and* as a class.